Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Types of electrodes

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
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Contents

1. 1 · Types of electrodes
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Types of electrodes

1 Types of electrodes

Study focus: Types of electrodes. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000026 · EV0000027

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module four on electrochemical equilibria and in today's class we are going to discuss about the different types of electrodes.

Key details

- The metal ion electrode, the metal insoluble salt ion electrode, the gas electrode, amalgam electrode, redox electrode and iron selective electrode.
- In order to form a cell, we need two half cells, and therefore two electrodes are required.
- The first one is a metal ion electrode.
- We have a metal rod which is in contact with a solution of its own ions.
- Remember that we always write the reduction reaction because we all talk about reduction potentials.
- Since a potential is formed at the.
- Electrode and the solution interface.
- This is a familiar Nant equation which we have already learnt.
- M of MN plus is equal to E naught that is the standard electrode potential of M Mn plus two point three zero three R T over NF log of Mn plus.
- So that is the first type of electrode.
- How would we describe this electrode?
- As the name suggests, we have a metal which is in contact with a sparingly soluble salt of the metal and that is again dipped in a solution of soluble salt of the same anion.
- Mercury, which is a metal, and that is in contact with a sparingly soluble salt of mercury, that is.
- A G2 Cl2 and now this is dipped in a solution of KCl.
- Because it should be the solution of a soluble salt of the same anion.

- And here what do we have in silver chloride electrode?
- We have the silver metal which is in contact with silver chloride and that is in contact with HCl.
- The equation is AG Cl plus electron is giving you AG plus Cl minus.
- What we have to be careful about is that this is not A G plus and even if we write A G plus the equation would not be balanced.
- Plus Cl minus and we can see this electrode over here in the diagram we have a silver wire which is coated with silver chloride that is the sparingly soluble salt of silver and that is dipped in a solution either of KCl or SCl.
- It should contain it should be a solution containing the anion and the anion of the salt is Cl minus.

Example or application from the lecture

- Let us see what would be the expression for the electrode potential, and that is given by the Nant equation.
- Let us come to the second type where we have metal, metal insoluble salt, iron electrode.
- The most common example for this would be the calomel electrode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 05:00–09:59

Zero point zero five nine one by one log of Cl minus.

Key details

- All of you are familiar with the hydrogen electrode.
- We know that in a hydrogen electrode you have this a bell-shaped jar which is being which is used here, and it has an inlet where we are going to let in hydrogen.
- It is usually platinum or it could be a platinum wire or a platinum foil.
- So and then this is now this whole assembly is dipped into a solution of H plus ions because those are the ions to which the hydrogen gas is reversible.
- The platinum wire which we are using, what is the purpose of that?
- It provides the electrical contact and facilitates the establishment of equilibrium between the gas and its ions.
- We have platinum in contact with H2.
- We have to.
- Use chlorine gas which is going to be bubbled around the platinum wire and now this has to be dipped in a solution of Cl minus ions.
- Electrical contact provid facilitates the establishment of equilibrium.
- between the gas and its ion.
- We are talking about this equilibrium.
- Now in the Nernst equation we have E is equal to E naught minus 0.0591 by 2 because here we are using two electrons.
- That's why we have to make sure we write the reduction reaction only.

- But hydrogen is a gas, so we have to use the partial pressure of hydrogen here and.
- What is in the reactant side?
- We have H plus ions.
- When we write the log term, it is partial pressure of hydrogen divided by H plus the concentration.
- Raise to the power of 2.
- So this log term goes to 0, and therefore we get E is equal to E naught.
- The fourth of electrode is an amalgam electrode.
- Now as the name suggested suggests, amalgam means any alloy of Merkle.
- metal metaline electrode the only difference is that the metal is now taken in the form of an amalgam.
- So therefore we have say if we have a lead amalgam, then we use lead in the form of an amalgam and it has to be dipped in a solution of its own ions.
- It is just like the.
- metal ion electrode where we have metal ion taking up electron and becoming the metal.
- But here the only difference is we are taking the metal in the form of an amalgam.

Example or application from the lecture

- Let us come to the third type of electrode that is the gas electrode.
- So hydrogen electrode is a good example for the gas electrode.
- So, example would be hydrogen electrode, as I mentioned just now.

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EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 10:00–14:59

Is very active.

Key details

- If we amalgamate the sodium, then we would have sufficiently reduced the activity of sodium.
- And then we can make this electrode.
- So not only that we can actually control the activity of this electrode because we can decide the percentage of mercury which we are going to add into the metal to the form the amalgam.
- 2 is the number of electrons log of.
- Whenever we have pure metal, the activity is taken as one.
- But here since we have amalgamated it.
- Definitely the activity will not be one.
- In fact it will depend on the percentage of lead which is there in this in this electron.
- Then we have redox electrodes, or they are also known as the oxidation reduction electrode.
- We have say tin.
- Which is in the form of two species Sn2 plus and the Sn four plus.

- When we have both these species present in the solution and we place a platinum wire inside, then we know.
- There is a natural tendency of one of the oxidation states to get converted into other, and in that process electrons are getting exchanged.
- formed and that elect that potential is being picked up by these by the inert electrode.
- The potential arises due to the tendency of one form to change into the other.
- And if we have both these species present in the same solution and we place a platinum.
- And wire there, we will get what we call as a redox electrode.
- Other examples would be platinum C3 plus C4 plus.
- or platinum Sn two plus S N four plus as you can see in the picture over here.
- It would be Sn four plus taking up electrons and forming S N two plus the Nernst equation will be based on this equation now.
- ELE is equal to E naught minus 0.0591 by 2 log of S N2 plus that is the products divided by the reactants.
- A special case of redox electrode is the quin hydron electrode.
- It consists of an inert metal such as platinum which is immersed in a solution containing quinone and hydroquinone.
- We can see here this is the reaction which is happening.
- It involves an exchange of electrons in the presence of H plus ion.
- Quinone and hydroquinone present together in the solution.
- And if we place a platinum electrode there, we will be able to get.

Example or application from the lecture

- So a good example of that would be your iron, which exists as Fe two plus and Fe three plus.

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EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 15:00–19:59

How do we represent Q H two and divided by that of quinone that is.

Key details

- Q concentration and we also have H plus.
- These are also referred to as membrane electrodes.
- An ion selective electrode, as the name suggests, it is going to select a particular ion in the presence of other ions.
- and it is going to tell us the concentration of that particular ion or in other term in other words it is going to develop a potential based on the concentration of that one particular ion.
- It is because it has a membrane which is in contact with the solution, and this membrane is the most important constituent, and this membrane is being able to exchange the ions and since it is being able to exchange the ions it can it develops a potential and that.
- That is why it can detect the concentration of one ion in the presence of other ions.
- And what are pH meters which tell us the concentration of electricity.
- H plus so the glass membrane is selective to H plus ion.

- Because it is selective to H plus ion, it can give us the concentration of H plus N, or in other words, it can help us determine the pH of a solution.
- And it is also sensitive to not only H plus, sometimes to NE plus and K plus.
- Pla E no E G would be equal to E naught G plus 0.0591 log of H plus Ion.
- Extremely sensitive and they are also known as membrane electrodes because membrane is the most important constituent of a glass electrode.
- And in the next class we are going to talk about reference electrodes.

Example or application from the lecture

- So, an example for this would be glass electrode, and we know that glass electrode is used in pH meters.

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EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Types of electrodes transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Types of electrodes transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

C Revision and retrieval practice